RIWA KARAM

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Irvine, CA

Professional Summary

PhD candidate in Electrical and Computer Engineering specializing in multi-agent systems, robotics, machine learning, and optimization. Experienced in developing data-driven models, designing collaborative robotic systems, and translating research into practical solutions. Seeking research opportunities that advance intelligent autonomous systems through rigorous analysis, experimentation, and effective teamwork.

Industry Experience

- **ASML** [] — *June 2025 – September 2025*
  Mechatronics Research Engineer Intern — Wilton, CT
  - Investigated machine health monitoring using statistical and Bayesian hierarchical models for uncertainty quantification and structural analysis.
  - Developed a Bayesian hierarchical model using Python and PyMC to predict drift and performance degradation, enabling root-cause analysis across machine lifetimes.
  - Built and compared decision tree, random forest, multiple regression, and baseline regression models to evaluate performance and identify patterns in the data.
  - Conducted data collection, preprocessing, and analysis using Python and MATLAB, including regression modeling, principal component analysis (PCA), and heatmap-based pattern recognition.
  - Applied Markov chain Monte Carlo (MCMC), Bayesian inference, and time-lag regression methods to characterize model uncertainty and temporal behavior.
- **ASML** [] — *June 2024 – September 2024*
  Mechatronics and Control Systems Engineer Intern — Wilton, CT
  - Developed a Gaussian process regression (GPR) model using Python and MATLAB to predict force generated by a piezoelectric actuator with accuracy within ± 2 N; the work contributed to an ASML invention.
  - Built regression and machine-learning models to identify correlations, dependencies, and performance drivers in experimental data.
  - Automated and merged MATLAB and Python scripts that run testing workflows for both the mechatronics and software teams, reducing runtime from 40 hours to an overnight run.
  - Investigated feasibility of optimization problems and current-control hardware issues for test stands by analyzing their electrical circuitry and alternative hardware.
- **AviaPro Consulting Inc.** [] — *June 2022 – July 2023*
  Software Developer — Remote
  - Developed web applications using React.js and Node.js, replacing Excel-based repositories with streamlined interfaces connected to multiple databases.
  - Applied software-development and DevOps practices, including CI/CD, infrastructure as code, and Git.
  - Contributed to team-based full-stack projects and independently developed an internal full-stack application.
  - Transitioned to a part-time role in September 2022 after completing a summer co-op.

Academic Experience

- **UCI Robot Ecology Laboratory** [] — *September 2023 - Present*
  Graduate Student Researcher — Irvine, CA
  - Conducting research in collaborative multi-agent systems, optimization, machine learning, ecology-inspired robotics, coverage control, formation control, and human-swarm interaction under the supervision of Professor Magnus Egerstedt and Professor Yanning Shen.
  - Maintaining the Robot Ecology Lab Robotarium testbed through robot debugging, repair, and system troubleshooting.
  - Presented a first-author regular paper at the 64th IEEE Conference on Decision and Control (CDC) in Rio de Janeiro, Brazil.
  - Presented research findings at the 45th Southern California Control Workshop (SCCW) at the University of California, San Diego.
  - Passed my PhD qualifying examination in May 2026 and advanced to PhD candidacy.
  - Won the Software & Algorithms track at the 2026 UCI GEECS Annual Technology Showcase.
  - Presented a poster on my master's thesis at the 2025 UCI GEECS Annual Technology Showcase.
  - Mentor undergraduate researchers and collaborate with them on related research projects.
  - Led laboratory Robotarium testbed tours and live multi-robot demonstrations for more than 100 visitors.
  - Co-authored the Robot Ecology Lab manual, documenting debugging procedures, experiment workflows, and Vicon system maintenance.

Education

- **University of California, Irvine** — *September 2023 - Expected June 2027*
  Electrical & Computer Engineering, Doctor of Philosophy — Irvine, CA
  - Dissertation: Collaboration in Multi-Agent Systems: From Assignment to Learning
  - Advisors: Professor Magnus Egerstedt, Professor Yanning Shen
- **University of California, Irvine** — *September 2023 - June 2025*
  Electrical & Computer Engineering, Master of Science — Irvine, CA
  - GPA: 3.975/4.00
  - Thesis: A Graphical Interface for Specifying and Establishing Multi-Robot Formations
  - Relevant Coursework: Optimization, Network Science, Control and Machine Learning, Linear Systems
- **University of Balamand, Al-Kurah** — *August 2020 - June 2023*
  Computer Engineering, Bachelor of Science — Balamand, Al-Kurah, Lebanon
  - GPA: 3.90/4.00
  - Computer Engineering Class of 2023 Valedictorian
  - Graduation Project: Gamification of Virtual Reality Kitchen Scenarios
  - Tutored peers through group and private sessions in courses such as programming (Python, MATLAB, C), Signals and Systems, Microcontrollers, Embedded Systems, and Computer Architecture.

Publications

C=Conference, T=Thesis, R=Under Review

[R.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Learning Altruistic Collaboration in Heterogeneous Multi-Team Systems**. Under Review in the IEEE Transactions on Robotics (T-RO). arXiv preprint version available, DOI.

[R.2] Riwa Karam, Alexander A. Nguyen, Ruoyu Lin, David R. Martin, Diana Morales, Brooks A. Butler, and Magnus Egerstedt, **Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration**. Under Review in the IEEE Control Systems Magazine (CSM). arXiv preprint version available, DOI.

[C.1] Riwa Karam, Ruoyu Lin, Brooks A. Butler, and Magnus Egerstedt, **Resource Allocation for Multi-Team Collaboration Based on Hamilton's Rule**, IEEE 64th Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025, pp. 6891-6898, DOI.

[T.1] Riwa Karam, **A Graphical Interface for Specifying and Establishing Multi-Robot Formations**, M.S. Thesis, University of California, Irvine, 2025, Official Record.

Skills

- **Programming Languages and Libraries:** Python, MATLAB, scikit-learn, PyTorch, Keras, NumPy, pandas, Matplotlib, shell scripting, PyMC, C, Java, C#.
- **Hardware Technologies:** Robot testbeds, Vicon Motion Capture (MoCap) System, robot building & debugging.
- **Other Tools & Technologies:** Git & GitHub, LaTeX, Simulink, Visual Studio Code, Unity, Microsoft Office, Google Workspace.

Selected Honors and Awards

- **GEECS Annual Technology Showcase First Place Winner** — *June 2026*
  Software & Algorithms Track, University of California, Irvine
- **Faculty Award for Academic Excellence** — *July 2023*
  University of Balamand, Faculty of Engineering
- **First Place Winner, Lebanon IoT & AI Challenge 2021** — *November 2021*
  Arab IoT & AI Challenge
- **Dean's Honor List** — *August 2020 - June 2023*
  University of Balamand, Faculty of Engineering

Professional Service

- **UCI Society of Graduate Electrical Engineers and Computer Scientists** — *June 2026 – Present*
  Web Content Manager — Irvine, CA
- **IEEE Control Systems Society NextCom** — *March 2026 – Present*
  General Activities Committee Member — Remote

Languages

- Arabic: Native proficiency
- English: Full professional proficiency
- French: Full professional proficiency
- Spanish: Elementary proficiency (CEFR A2)